

Paediatric Headache

HEADLINE NEWS:

- Most parents worry about headache being a brain tumour symptom in their child.
- Most headache is not brain tumour
- Most headache responds to lifestyle improvements
- Prevalence of brain tumour is much lower (1:100,000) than the prevalence of tension or migraine headaches (1:15!).
- Useful to use the Headsmart decision making tool during your consultation to reassure parents you are considering brain tumour! Headsmart was a campaign set up to reduce the time taken to diagnose a brain tumour in a child. It is endorsed by the RCPH.

Please also see helpful resource on headache:

[Infographics #headache - PEM Infographics](#) and [Headaches Module - Don't Forget the Bubbles](#)

Useful Leaflets for Parents

[Migraine in children](#)

[Prophylactic treatment of migraine in children](#)

Flow chart for Primary Care Management of Headache in Children in Derbyshire



Diagnosis of tension-type headache, migraine and cluster headache

Headache feature	Tension-type headache		Migraine (with or without aura)		Cluster headache	
Pain location ¹	Bilateral		Unilateral or bilateral		Unilateral (around the eye, above the eye and along the side of the head/face)	
Pain quality	Pressing/tightening (non-pulsating)		Pulsating (throbbing or banging in young people aged 12–17 years)		Variable (can be sharp, boring, burning, throbbing or tightening)	
Pain intensity	Mild or moderate		Moderate or severe		Severe or very severe	
Effect on activities	Not aggravated by routine activities of daily living		Aggravated by, or causes avoidance of, routine activities of daily living		Restlessness or agitation	
Other symptoms	None		- Unusual sensitivity to light and/or sound or nausea and/or vomiting. - Aura: symptoms can occur with or without headache and; are fully reversible, develop over at least 5 minutes, last 5 - 60 minutes. Typical aura symptoms include visual symptoms such as flickering lights, spots or lines and/or partial loss of vision; sensory symptoms such as numbness and/or pins and needles; and/or speech disturbance.		On the same side as the headache: - Red and/or watery eye - Nasal congestion and/or runny nose - Swollen eyelid - Forehead and facial sweating - Constricted pupil and/or drooping eyelid	
Duration of headache	30 minutes–continuous		4–72 hours in adults 1–72 hours in young people aged 12–17 years		15–180 minutes	
Frequency of headache	< 15 days per month	≥ 15 days per month for more than 3 months	< 15 days per month	≥ 15 days per month for more than 3 months	1 every other day to 8 per day ³ , with remission ⁴ >1 month	1 every other day to 8 per day ³ with a continuous remission ⁴ <1 month in a 12-month period
Diagnosis	Episodic tension-type headache	Chronic tension-type headache ²	Episodic migraine (with or without aura)	Chronic migraine (with or without aura)	Episodic cluster headache	Chronic cluster headache

¹ Headache pain can be felt in the head, face or neck. ² Chronic migraine and chronic tension-type headache commonly overlap. If there are any features of migraine, diagnose chronic migraine. ³ Frequency of recurrent headaches during a cluster headache bout. ⁴ The pain-free period between cluster headache bouts.

Adapted from 'Headaches' (NICE clinical guideline 150), available from www.nice.org.uk/CG150

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Management

- **Explanation and reassurance.** This should emphasise what is causing the headache and how realistically attacks can be reduced. Fears about brain tumours and more serious conditions should be discussed and fears allayed.
- **Identification of triggers and predisposing factors** with a trigger/headache diary. Record frequency, duration and severity of headaches. This may be important in the development of a behavioural strategy. Dietary sensitivities affect only about 20% of migraine sufferers. In children, the following triggers have been identified as important: sleep, stress, dehydration/warm weather, missed meals, video games.
- **Behavioural management strategy.** This should emphasise routine around sleeping, eating and avoiding an overloaded routine to help prevent migraine, as well as strategies for dealing with an attack (lie in a cool, dark, quiet room and encourage sleep with pharmacological or non-pharmacological support).

Type	Treatment	Other info
1. Tension-Type headache	Paracetamol / NSAID for acute attacks First line NSAID Ibuprofen	
2. Cluster headache	Discuss need for neuroimaging via A&G / CC Offer nasal triptan for acute treatment +/- oxygen	
3. Medication overuse headache	Stop all analgesia for at least 1 month Consider prophylactic treatment for underlying primary headache disorder	
4. Migraine (Common! Present in 10% children)	Acute attack: First line - analgesic (Paracetamol / NSAID) with antiemetic e.g., Domperidone, Cyclizine Second line Age >12 consider use of Nasal Triptan e.g., Sumatriptan Initially 10–20 mg for 1 dose, followed by 10–20 mg after at least 2 hours if required, to be taken only if migraine recurs (patient not responding to initial dose should not take second dose for same attack); maximum 40 mg per day Age <12 Consider use of Oral Triptan after discussion via A &G or CC if unsure. Aged 10-11 years initially 50mg for 1 dose, followed by 50mg after a least 2 hours, to be taken only if the migraine recurs.	Domperidone - MHRA Dec 2019 Domperidone is no longer licensed for use in children <12 years or those weighing <35 kg. Results from a placebo-controlled study in children younger than 12 years with acute gastroenteritis did not show any difference in efficacy at relieving nausea and vomiting compared with placebo. MHRA Dec2014 Risks of cardiac side effects - For adults and children ≥12 years old and ≥ 35 kg, the recommended max. dose is 30mg in 24h (10mg up to TDS) - In children <12 years old weighing <35 kg, recommended max. dose is 0.75 mg/kg in 24h (0.25 mg/kg up to TDS) Max. duration should not usually exceed 1 week Cyclizine - Unlicensed for children under 6 For child 6-11 years oral dose 25mg up to TDS For child 12-17 years, oral dose 50mg up to TDS Sumatriptan - Oral sumatriptan unlicensed in children <18years of age but recommended by NICE. Nasal triptan licensed in children >12 years of age BNFC dosing info for treatment for acute migraine Oral 6-9 yrs.- 25mg, 10-11 yrs.- 50mg, 12-17 yrs.- 50-100mg Intranasal 12-17 years- 10-20mg (max 40mg per day)
5. Migraine Prophylaxis:	Consider if frequent and affecting school attendance Topiramate (beware in girls of childbearing potential), Propanolol Prophylaxis with dosing as per BNFC	Topiramate - Not licensed for use in children for migraine prophylaxis but recommended by NICE. BNFC- Child 16-17 yrs.- 25mg ON for 1wk, then ↑ in steps of 25 mg every week; usual dose 50–100 mg daily in 2 divided doses; maximum 200 mg per day. MHRA Jan21 An increased risk of major congenital malformations and fetal growth restriction associated with topiramate.

Flow chart for Primary Care Management of Headache in Children in Derbyshire

		Propanolol- BNFC- Child 2-11 yrs. initially 200-500mcg/ kg BD; usual dose 10-20mg BD. Max. 2mg/kg BD Child 12-17 yrs. initially 20-40mg BD; usual dose 40-80mg BD Max. per dose 120mg; Max. 4mg/kg per day Severe overdoses with propranolol may cause cardiovascular collapse, CNS depression, and convulsions. People with depression and migraine could be at an increased risk of using propranolol for self-harm. Use caution when prescribing propranolol, in line with the Healthcare Safety Investigation Branch's report on the under-recognised risk of harm from propranolol
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Drugs recommended in this pathway have been checked against NICE, BNFC, and Derbyshire formulary.

Referral Criteria for Children with Migraine:

- If red flags, please refer as headache pathway above
- Migraine has been present over 72 hours – please refer via Consultant Connect / Consultant on-call at CRH
- Need for second line treatment
- Uncertainty about diagnosis
- Please refer via usual Outpatient route

For more detail, please consider NICE guideline:
<https://www.nice.org.uk/Guidance/CG150>